PS 170A Experiment Final Report

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Part 1: Introduction

The behavioral problem is overuse of social media. We observed that college students devote a large portion of their time to social media apps and thus, we feel that they would benefit from reducing their screen time.

To solve this issue, we focus on two secondary outcomes: targeting the potential lack of knowledge on how to reduce screen time and targeting the beliefs/norms people of how social media impacts the body and mind. We would see which secondary outcome addresses the behavioral problem and reduces social media use the most effectively by self-reporting through a survey after watching knolwedge-targeting and belief targeting videos. We would ask for screenshots of their screen time before and after treatment.

Our idea works because it measures and changes the behavior of participants' screen times by attempting to target the secondary outcomes on why people do not take on the prosocial behavior on reducing their social media consumption. People may not know how to reduce their screen time or believe it will not lead to any change in their lives; thus, we develop an intervention that surveys those secondary outcomes. For reliability, we will ask for screenshots before and after treatment.

Part 2: Treatment

We randomly assign one of two conditions, in which one condition is the treatment that targets the knowledge and beliefs of the participant while the other is the control that has no intervention.

The treatment will include three videos: one knowledge-targeting video (6 Ways to Reduce Screen Time: https://www.youtube.com/watch?v=XQQUo2gmUs4) and two belief targeting videos (Why our screens make us less happy | Adam Alter: https://www.youtube.com/watch?v=XQQUo2gmUs4 + What Staring At A Screen All Day Is Doing To Your Brain And Body | The Human Body: https://www.youtube.com/watch?v=rx1l51qxGbQ&ab_channel=InsiderTech)

We will schedule to send a first email on 3/15 in the afternoon, asking them to self-report their time spent on entertainment social media apps and include screenshots to be held accountable. We will ask them to then report time on entertainment apps later to our treatment email (3/18 - 3/20) to see the differences in screen time between the two. In our email, we want to incentivize people to fill out the surveys by emphasizing that this is a class project and including a $15 gift card raffle.

CONTROL EMAIL: Dear students of UCLA,

As the quarter comes to an end, we find ourselves wanting to finish final projects. As fellow students, we come to you in need of data for our final project, which also comes with the chance of winning a $15 gift card of your choosing. As students enrolled in a statistical analysis course, we have been tasked with formulating a project that measures a variable of our choosing. In our project, we wish to measure the total minutes of screen time students spend on social media, such as: Twitter, Instagram, TikTok, and Snapchat. All we ask of our fellow UCLA students is that

they fill out the survey attached to this email. In this survey, you will be asked to self-report the daily average screen-time you have spent on these apps (which is the entertainment section of screen-time on an iPhone). Additionally, you will be asked to attach a screenshot of your official recorded screen time for the week (this can be found under settings in an iPhone and Android). Once this survey has been completed, your name will be entered into a raffle where you will have the chance to win a $15 gift card of your choosing.

Survey link: https://docs.google.com/forms/d/e/1FAIpQLSf07bBlcBmCBcL2QW4oVo1nLgZCnDsGb05TE4JdhNR-yGZGnQ/viewform

We thank you for your time and patience in completing this survey.

TREATMENT EMAIL: Dear students of UCLA,

As the quarter comes to an end, we find ourselves wanting to finish final projects. As fellow students, we come to you in need of data for our final project, which also comes with the chance of winning a $15 gift card of your choosing. As students enrolled in a statistical analysis course, we have been tasked with formulating a project that measures a variable of our choosing. In our project, we wish to measure the total minutes of screen time students spend on social media, such as: Twitter, Instagram, TikTok, and Snapchat. Social media has created negative consequences among our generation (imposter syndrome, interpersonal comparison, etc.) Its content and use has been found to significantly impact the mental health of users, and to have addictive qualities that hinder productivity, thus further worsening anxiety and other mental health factors of students. This experiment aims to find the most productive way to reduce screen time of college students, which is important because reduced screen time would inherently reduce social media usage, and therefore hopefully reduce the stress and improve productivity in students. To further see the negative effects of social media and why screen time should be limited, look at the sources attached below: https://www.youtube.com/watch?v=0K5OO2ybueM&ab_channel=TED and https://www.youtube.com/watch?v=rx1l51qxGbQ&ab_channel=InsiderTech

All we ask of our fellow UCLA students is that they fill out a survey that will be sent to their email on 3/15. In this survey, you will be asked to self-report the daily average screen-time you have spent on these apps (which is the entertainment section of screen-time on an iPhone). Additionally, you will be asked to attach a screenshot of your official recorded screen time for the week (this can be found under settings in an iPhone and Android). Once this survey has been completed, your name will be entered into a raffle where you will have the chance to win a $15 gift card of your choosing.

Survey link: https://docs.google.com/forms/d/e/1FAIpQLSf07bBlcBmCBcL2QW4oVo1nLgZCnDsGb05TE4JdhNR-yGZGnQ/viewform

We thank you for your time and patience, and look forward to sending out the survey at the end of this week.

FOLLOW-UP EMAIL: At the beginning of this week, we sent out an email detailing our student-led project, attempting to record screen time among UCLA students. As the quarter comes to an end, we find ourselves wanting to gather the data needed for this project, and now ask that you fill out the survey attached to this email. When you have completed this survey, your name will be entered in a raffle where you will have the chance of winning a $15 gift card of your choosing. Survey link: https://docs.google.com/forms/d/e/1FAIpQLSf07bBlcBmCBcL2QW4oVo1nLgZCnDsGb05TE4JdhNR-yGZGnQ/viewform. Once again, all we ask of our fellow UCLA students is that they fill out the survey attached to this email. In this survey, you will be asked to self-report the daily average screen-time you have spent on these apps (which is the entertainment section of screen-time on an iPhone). Additionally, you will be asked to attach a screenshot of your official recorded screen time for the week (this can be found under settings in an iPhone and Android). Once this survey has been completed, your name will be entered into a raffle where you will have the chance to win a $15 gift card of your choosing. We thank you for your time and patience in completing this survey.

Part 3: Outcomes

Our main outcome is measuring the participant's daily average screen-time on their social media apps. We worded the question: How many minutes per day do you spend on social media or entertainment apps (Instagram, TikTok, Twitter, etc.)? We've ensure the results are valid and reliable by also having them to include a screenshot of their screen time.

Our first secondary outcome is the fear of missing out, where social media has become a norm for people to keep up with what their friends are doing, topics that are trending, and conform to what others are doing. We worded the question: On a scale of 1-5, 1 being "not at all" and 5 being "completely", how much do you think your social media use is driven by the fear of missing out?

Second secondary outcome is the belief that monitoring their social media consumption would not have a positive outcome. We worded the question: Do you correlate screen-time with productivity? (1 for 'yes' or 0 for 'no')

Our last secondary outcome is the lack of knowledge on the consequences of too much social media usage. We measure this with two questions: Give an estimated time for how many hours of social media use per day will lead to individuals being at risk for mental health struggles (scale of 0-3 hours to 10+ hours). Adolescents would be at risk for mental health struggles if they use social media for more than 3 hours per day (0 for 'no' and 1 for 'yes').

Part 4: Research design

Our research questions is: Which secondary outcome, when targeted, reduces social media screen time the most effectively? To answer this question, we designed a experiment that will send out a control and treatment email, with the control email being similar to the treatment but will include no invervention (no videos). It will only explain the experiment and its purpose.

Our outcome of interest, dependent variable, is the entertainment screen time measured, on average, minutes that each individual spent on entertainment apps in the week prior to the treatment email and the week after. Our outcome will be measured through a survey asking bout social media behavior on various apps such as: Snapchat, Twitter, Tik Tok, and Instagram. The survey will have individuals self-report on the daily average screen-time people use on these various apps. To ensure these results are valid, we could ask participants to also include a screenshot.

To measure the fear of missing out secondary outcome, we will ask: How much do you think your social media use is driven by the fear of missing out? The answers will be a scale of 1-5, 1 being "not at all" and 5 being "completely". To measure beliefs secondary outcome, we will ask Do you correlate screen-time with productivity? This question will have a binary answer of either 1 for "yes" or 0 for "no". We will also ask participants if they feel that after going through the treatment and monitoring screen-time, there was an effect on their productivity. This will be based on a scale of 1-5, 1 being "no effect at all" and 5 being "extremely significant effect". To measure lack of knowledge, we will ask a question to demonstrate if they believe there is a threshold above which social media use would begin to put adolescents at risk for mental health problems. According to a study done by JAMA Psychiatry, over three hours of social media use per day leads to an increased risk in mental health issues (Riehm et al 2019). Therefore, we will use 3 as our baseline number to gauge their knowledge. This question will be formatted as: Adolescents would be at risk for mental health struggles if they use social media for more than 3 hours per day. The only answers possible will be "Yes" (coded as 1) and "No" (coded as 0). We will subsequently ask where participants would place the threshold (in terms of hours). Our question will be: Give an estimated time for how many hours of social media use per day will lead to individuals being at risk for mental health struggles. The answers will be 4 options: 0-3, 4-6, 7-9, 10+ (coded as a scale from 1-4), allowing us to understand their baseline knowledge on the impact of social media usage.

There are outcome interests we can't measure: usage per individual apps and mental health. We are interested in how usage may change per apps; indicating if a feature of one app or apps are the cause of prolonged social use, its best to reduce the screen time of that app. We are also not sure on how to measure mental health through a survey.

To visualize the raw data, we will use the ggplot function to create a scatter plot on the various differences in screen-time across treatment and control.

Part 5: Analysis

Analyses and visualizations

```
social_media <- read.csv("https://www.dropbox.com/s/fzvismasj3lkwil/sectionf_final_data.csv?dl=1")
```

Codebook:

- `treatment`: "treatment"
- `minutes`: "time after treatment"
- `fomo`:"How much do you think your social media use is driven by the fear of missing out?"
- `productivity`:"Do you correlate screen-time with productivity?"
- `mental_health`: "Give an estimated time for how many hours of social media use per day will lead to individuals being at risk for mental health struggles"
- `adolescent`: "Adolescents would be at risk for mental health struggles if they use social media for more than 3 hours per day."

Clean data

```
#colnames renames the variables
colnames(social_media) [2]='minutes'
colnames(social_media) [4]='fomo'
colnames(social_media) [5]='productivity'
colnames(social_media) [6]='mental_health'
colnames(social_media) [7]='adolescent'

#change minutes into hours
social_media <- social_media %>%
  mutate(hours = case_when(minutes == "3.3 hours" ~ 3.3,
                           minutes == "258" ~ 4.3,
                           minutes == "7 hours 12 minute average" ~ 7.2))
#delete minutes variable
social_media <- subset(social_media, select= -minutes)
```

Estimated Treatment Effect Size

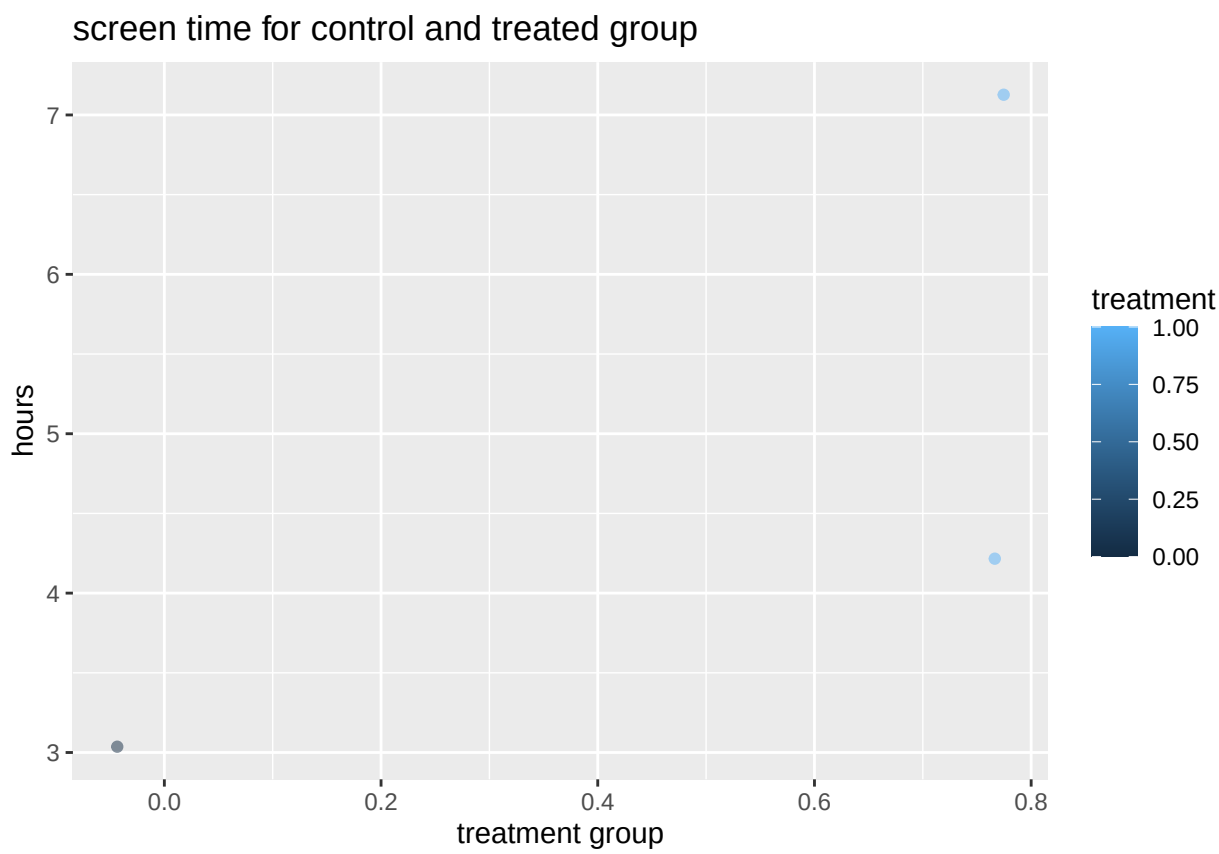
```
# difference in means between the mean of treatment and control group
# effect_size <- difference_in_means(hours ~ treatment, data = social_media) %>%
#  tidy
# print(effect_size)
social_media %>%
  summarise(dim = mean(hours[treatment==1], na.rm = TRUE) -
                              mean(hours[treatment==0], na.rm = TRUE))
```

```
##    dim
## 1 2.45
```

Based on three units and attempting to calculate the estimated treatment effect, we have gotten 2.45 for difference in means between the treatment and control group. This is not an accurate calculation of average treatment effect due to the sample size only being three individuals, making it difficult to calculate the difference in means. We were unable to calculate the treatment effect size due to the 'design is not pair-matched', where there must be least two treated and control units. Thus, the p-value cannot be calculated to determine if we are confident in our treated estimate. Along with the small sample size of three units representing pre-treatment groups, we cannot make a firm conclusion whether this experiment has been statistically significant.

Visualize data

```
# ggplot to graph the control and treated groups on x axis and
# the difference in times in terms of hours on y axis
ggplot(data = social_media, aes(x = treatment, y = hours, color = treatment)) +
  #geom_jitter creates scatter plot
  geom_jitter(alpha = 0.5, height = 0.3, width = 0.3) +
  # labs creates titles
  labs(x = "treatment group", y = "hours",
       title = "screen time for control and treated group")
```



The scatter plot displays the hours of the control group and the treated group of both knowledge and beliefs. The graph shows the range of hours between the control and treatment, with the treatment group having more screen time hours than the control group.

Results

We changed the minutes to hours in order to make the units the same across board. Nobody had taken the follow up treatment email, so we couldn't calculate the difference in the times to analyze the effectiveness of intervention. With incomplete data as well as limited sample size, we were unable to calcualte the difference between treated and control group and operate a power analysis experiment (mentioned in our pre-analysis plan). Therefore, we had to analyze the data based on pre-treatment groups of one unit in control and two in treatment.

The code changed for difference in means. The built in DIM function would not take one control unit, so we had to use the summarize function to calculate the mean of each control and treat group. We were unable to code the power analysis experiment due to a very small sample size of three units, so we could not calcuate the standard deviation for the control potential outcome.

We had to change the analyses of the estimated treatment size and scatter plot to focus on the pre-treatments, rather than analyzing if the intervention had worked or not. Since no one had follow up with the email, we cannot determine whether this intervention is effective.

The results were unexpected since we received a limited random sample size and no data for the hours recorded after the treatment. If considering the results found in pre-analysis plan, we would have planned to interpret whether this treatment is effective in decreasing screen time after watching a video intervention in the email. We would see if there is a statistically significant difference between control and treated group, where the control group would be less likely to reduce their screen time compare to the treated group after the intervention. We would also see if the experiment is sufficiently powered by running a power thought experiment, ensuring that our sample size was adequate for providing a statistically significant result.

Part 6: Conclusion

Our experiment is about the behavioral problem on social media overuse, and we target two secondary outcomes to find which optimally addresses our behavioral problem: the potential lack of knowledge on how to reduce screen time or targeting the beliefs people may have on how social media impacts the body and mind. A prolonged exposure to social media can cause issues of anxiety and depression from the lack of productivity or self-consciousness of others. Due to the negative mental health effects, the goal of our experiment is to try to find the most effective way to reduce screen time. We will have one treatment and control email, with the treatment targeting knowledge and beliefs and control having no intervention. We have found that three people had answered to our survey and the intervention was not fully completed with a follow up email, leaving incomplete data. Since we cannot fully analyze our given data to its potential, we can only assume to adopt neither video treatments. We have learned to recognize some of the barriers inherent to this kind of project, so we would have done the experiment differently if we were not restrained in time. This includes revising the emails by making two treatment emails and one control email that are more concise, a longer time frame to send the treatment email and a follow up email, and correcting the surveys into having the ability to input the participants' screenshots of their screentime.

Works Cited

- https://www.tutorialspoint.com/how-to-remove-a-column-from-an-r-data-frame
- https://jamanetwork.com/journals/jamapsychiatry/fullarticle/2749480?guestAccessKey=866a0678-6da9-4177-9c61-da0accc8257c&utm_campaign=ftm_links&utm_content=tfl&utm_medium=referral&utm_source=For_The_Media&utm_term=091119